# Technical Assessment of Ratio-Preserving and Shape-Preserving Warps for Real-Time Six-Camera 360-Degree Panoramic Imaging

Relevance, Evidence, Projection Geometry, and FPGA Deployability

Technical Evaluation Report

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**Abstract**

This report evaluates whether two 2018 natural-image stitching methods provide a technically valid alternative to the calibrated cylindrical projection used in a real-time, six-camera, 360-degree electro-optical and infrared panoramic system. The analysis first resolves a bibliographic conflation between *Ratio-Preserving Half-Cylindrical Warps for Natural Image Stitching* and the supplied paper *A Natural Shape-Preserving Stereoscopic Image Stitching*. It then examines citation history, evidence of adoption, algorithmic assumptions, measured runtime, projection geometry, compatibility with a deterministic FPGA streaming architecture, and a customer-proposed local-overlap warp variant. The central geometric result is that a rectangular pinhole sensor does not observe a constant vertical elevation range across its horizontal field of view. For a camera sector of half-angle $\alpha$ and vertical half-field $\beta$, the available elevation at azimuth $\theta$ is bounded by $\phi_{\max}(\theta) = \tan^{-1}(\tan\beta\cos\theta)$. No two-dimensional warp can recover rays outside that bound. The reviewed methods improve visual naturalness by scene-dependent rescaling and piecewise warping, but they do not preserve angular geometry or recover vertical field of view. Test-case data show that confining the remaining correction to narrow overlaps creates high-gain seam deformation rather than a calibrated projection. The report concludes that these methods may be useful for optional non-metric display rectangling, but they do not substantiate replacing calibrated cylindrical reprojection in the target real-time system.


**Keywords:** panoramic imaging, cylindrical projection, image stitching, vertical field of view, FPGA, multi-camera systems, real-time video, stereoscopic stitching.

## 1 Scope and Evaluation Question

The engineering question is not whether an adaptive warp can make a pair of still images look more rectangular. It is whether such a warp can simultaneously satisfy the following system requirements:

- calibrated ray geometry over a complete 360-degree azimuth;
- six fixed cameras with narrow, wide-baseline usable overlaps;
- sustained frame-synchronous video operation at 30 frames/s;
- deterministic execution on a fixed Xilinx XCKU15P FPGA platform;
- electro-optical (EO) and infrared (IR) operation without scene-dependent feature failure;
- inertial roll/pitch stabilization before seam formation;
- bounded line-buffer memory and sequential DDR transactions; and
- stable output geometry suitable for surveillance, measurement, and downstream analytics.

The EO panoramic content is $3840 \times 480$ and is transported by folding it into two $1920 \times 480$ bands in the HD frame. This transport fold does not alter projection geometry. The IR panorama has a separate $1788 \times 480$ active geometry and is not used in the EO field-of-view derivation below.

The appropriate test is therefore a requirements and evidence comparison, not a subjective comparison of selected stitched photographs.

# 2 Bibliographic Disambiguation and Citation Audit

## 2.1 Two distinct papers were conflated

The supplied file `wang2018.pdf` is Wang *et al.*, *A Natural Shape-Preserving Stereoscopic Image Stitching*, published at ICASSP 2018 [2]. It is a five-page conference paper about the consistency and viewing comfort of two stitched stereoscopic image pairs. It is not a half-cylindrical-warp paper.

The named work, Xu *et al.*, *Ratio-Preserving Half-Cylindrical Warps for Natural Image Stitching*, is a three-page arXiv preprint [1]. It was not published in *IEEE Transactions on Multimedia*. The likely source of confusion is that its references include the separate work *Quasi-Homography Warps in Image Stitching*, which was published in an IEEE transactions venue.

**Table 1:** Bibliographic and citation status as checked on 12 August 2026. Counts differ slightly across indexes because their coverage and deduplication policies differ.

| Item | Publication status | Citation record | Public adoption evidence |
|---|---|---|---|
| Xu *et al.* (RPHCW) | Three-page arXiv preprint; arXiv:1803.06655 | OpenAlex: 1; ResearchGate: 1 | No public code, product, standard, FPGA/GPU implementation, or replicated deployment found |
| Wang *et al.* (stereoscopic warp) | ICASSP 2018 conference paper; full identifier in Ref. [2] | OpenAlex: 4; Semantic Scholar: 3; ResearchGate: 3 | Later academic related-work citations; no public real-time 360-degree hardware adoption found |

## 2.2 Citation does not imply adoption

The one indexed citation to Xu *et al.* is Kang *et al.* [3], a 2019 cylindrical panorama paper for low-texture scenes. That paper mentions the half-cylindrical method as related work on global image quality; it develops a different alignment pipeline and does not implement or validate RPHCW.

The small citation lineage of Wang *et al.* remains within stereoscopic or natural-image warping. Zhang *et al.* [4] cite it before introducing a different global formulation with rectangular-boundary constraints. Yan *et al.* [5] describe it as an SPHP-derived transition warp and identify accumulated rotation and scaling in multi-image stitching. Chen *et al.* [6] cite it in the broader natural-stitching literature, while Nam and Han [7] address harmonization of stereoscopic 360-degree images using a different model.

These references establish that the papers were noticed in their narrow academic domain. They do not establish that either method was adopted as a calibrated, real-time, metric, multi-camera projection engine. Citation count itself is not a correctness test; the decisive issue is whether the cited work tests the same problem under comparable constraints.

# 3 Technical Reconstruction of the Reviewed Methods

## 3.1 Ratio-preserving half-cylindrical warp

The RPHCW pipeline [1] is a scene-dependent, two-image still-stitching method:

1. SIFT features are extracted and matched.
2. RANSAC estimates a global homography $H$.
3. The target image is partitioned into overlap and non-overlap regions.
4. The overlap is warped by $H$ to retain feature alignment.
5. The non-overlap is warped by a composition $C \circ H$, where $C$ is a cylindrical warp.
6. A similarity transform estimated from image features determines a horizontal resampling scale.
7. A seam is estimated and the images are blended.

The algorithm is explicitly piecewise: one region is projective, while another is projective-plus-cylindrical and then horizontally resampled. Consequently, the final image is not a single calibrated projection surface. Pixel spacing does not have one globally consistent angular interpretation.

The focal parameter is selected by minimizing an average-height error after warping. Horizontal samples are then reassigned approximately one output pixel apart according to a similarity scale. These operations preserve a chosen visual ratio; they do not preserve the original set of rays or a constant number of pixels per degree.

## 3.2 Measured runtime in the half-cylindrical preprint

Table 2 reproduces the reported runtime evidence. Seam processing is accelerated by resizing the image to one-eighth resolution and mapping the resulting seam back to the original image.

**Table 2:** Runtime reported by Xu *et al.* [1]. Frames/s is the reciprocal of the reported post-resizing total time and applies to one image pair.

| Resolution | Warp time (s) | Original total (s) | Reduced total (s) | Equivalent pair rate |
|---|---|---|---|---|
| $1500 \times 2000$ | 0.09 | 59.71 | 1.43 | 0.70 frames/s |
| $1920 \times 2560$ | 0.16 | 127.13 | 2.18 | 0.46 frames/s |
| $2448 \times 3264$ | 0.26 | 195.47 | 3.65 | 0.27 frames/s |

The paper does not identify the execution hardware for this table, report a six-camera schedule, or demonstrate video continuity. Even the fastest reported total is more than forty times the complete 33.3 ms budget for a 30 frames/s system, before processing all camera boundaries. The warp kernel alone reaches 0.09 s at the smallest resolution, which already exceeds the frame budget by a factor of 2.7.

The final sampling warp could, in principle, be computed offline and encoded into a lookup table. That observation does not make it a suitable calibration map: the map depends on features, a scene-derived homography, a scene-derived similarity, and a seam selected from one image pair. Baking such a result into hardware would encode the geometry of a calibration scene, not the fixed ray geometry of the camera rig.

## 3.3 Natural shape-preserving stereoscopic stitching

Wang *et al.* [2] solve a different problem. Their inputs are two stereo pairs, each containing left and right views. The method seeks low vertical disparity and visually comfortable stereoscopic output through:

- three sets of left/right feature correspondences;
- SIFT, RANSAC, moving-DLT/APAP local homographies;
- a stated 2000-iteration search for low-energy local transformations;
- Hough-transform straight-line extraction;
- a mesh energy containing alignment, smoothness, line, and disparity terms; and
- alpha blending of the final left and right panoramas.

The paper contains no cylindrical reprojection, no 360-degree closure, and no video experiment. Its evaluation comprises five stereo still-image cases from an existing dataset and one primary numerical metric: average absolute vertical disparity calculated at matched features. It reports no execution time, memory, latency, power, hardware mapping, temporal stability, angular fidelity, retained field of view, or infrared test.

The figures show substantial visual overlap, but the paper does not quantify overlap percentage or baseline geometry. It is therefore more rigorous to describe the test overlap as *unquantified and visually generous*, rather than assign an unsupported numerical value.

# 4 Requirements Comparison

**Table 3:** Comparison against the target real-time panoramic system.

| Requirement | Xu *et al.* RPHCW | Wang *et al.* stereoscopic warp | Target six-camera system |
|---|---|---|---|
| Input organization | One ordinary image pair | Two left/right stereo pairs | Six synchronized EO or IR cameras |
| Primary objective | Natural-looking still mosaic | Stereo disparity comfort | Calibrated 360-degree geometry and stabilization |
| Alignment source | Runtime SIFT/RANSAC | Runtime SIFT/RANSAC/APAP | Offline calibration plus frame-synchronous IMU |
| Projection surface | Piecewise $H$ and $C \circ H$ | Projective/shape mesh; no cylinder | One shared calibrated cylinder |
| Angular metric | Not preserved | Not defined | Stable inverse map from panorama pixel to source ray |
| Temporal evidence | None | None | 30 frames/s streaming modes |
| 360-degree closure | Not evaluated | Not evaluated | Required across all six camera sectors |
| Infrared operation | Not evaluated | Not evaluated | Dedicated IR geometry and runtime mode |
| Runtime determinism | Feature- and seam-dependent | Iterative, feature- and line-dependent | Bounded fixed-point schedule and line caches |
| Hardware evidence | None | None | Routed XCKU15P bitstream and live captures |

For distant content, the user's geometric intuition is substantially correct. When scene depth tends to infinity, translation-induced parallax tends to zero, and a homography is adequate for local overlap alignment under rotational camera motion [8]. Cylindrical projection is still useful for defining a common wide-angle or 360-degree coordinate system, but it is not needed to rescue an already nearly planar distant overlap. In both reviewed papers, the sophisticated non-overlap warp primarily controls visual projective distortion; it is not evidence of improved metric 360-degree geometry.

# 5 Why a Geometrically True Rectangle Loses Vertical Field of View

## 5.1 Pinhole-ray derivation

Consider a camera whose optical axis defines local azimuth zero. Let $\theta$ be a ray's local horizontal angle and $\phi$ its elevation. A unit ray may be written as

$$\mathbf{r}(\theta, \phi) = \begin{bmatrix} \cos\phi \sin\theta & \sin\phi & \cos\phi\cos\theta \end{bmatrix}^T . \tag{1}$$

Under pinhole projection, the normalized source coordinates are

$$x_n = \frac{r_x}{r_z} = \tan\theta, \qquad y_n = \frac{r_y}{r_z} = \frac{\tan\phi}{\cos\theta}. \tag{2}$$

Let the sensor's vertical half-field be $\beta$, so that $|y_n| \le \tan\beta$. Combining this with Eq. (2) gives

$$|\tan\phi| \le \tan\beta\cos\theta, \qquad \phi_{\max}(\theta) = \tan^{-1}(\tan\beta\cos\theta) . \tag{3}$$

The valid top and bottom boundaries therefore curve inward as $|\theta|$ approaches the camera-sector edge. This is not a defect introduced by cylindrical projection. It is the ray footprint captured by a rectangular pinhole sensor.

To retain a desired vertical half-field $\beta_d$ across a sector with horizontal half-width $\alpha$, the source optics must satisfy

$$\beta_{\text{source}} \ge \tan^{-1}\left(\frac{\tan\beta_d}{\cos\alpha}\right) . \tag{4}$$

No image warp changes this inequality. A warp may crop, leave invalid pixels, duplicate border samples, interpolate, or nonuniformly compress the captured rays, but it cannot create the missing elevations.

### 5.2 Numerical EO example

The EO camera specification used in the internal manuscript gives a vertical field of view of 51.3°, hence $\beta = 25.65°$. For a six-camera rig with nominal 60° sectors, $\alpha = 30°$. Equation (3) gives

$$2\phi_{\max}(30°) = 2\tan^{-1}(\tan 25.65° \cos 30°) \approx 45.16°. \tag{5}$$

Thus the sensor supplies the full 51.3° vertically near each camera's center, but only about 45.2° of elevation at the edges of a 60° sector. Preserving 51.3° everywhere in a rectangular metric projection would require

$$2\beta_{\text{source}} \approx 2\tan^{-1}\left(\frac{\tan 25.65°}{\cos 30°}\right) \approx 58.0°. \tag{6}$$

The angular-validity ratio $45.16/51.3 \approx 0.880$ is close to the current EO panorama crop scale near 0.888. The precise implementation value depends on calibrated intrinsics, distortion, the virtual cylinder parameterization, and discrete map boundaries, but its magnitude follows directly from the geometry.

### 5.3 What the reviewed warps actually preserve

The half-cylindrical preprint chooses a desired average output height, solves for a focal parameter, and resamples horizontal samples according to a scene-derived similarity. This can make objects appear less stretched, but it changes the relationship between output position and incoming ray direction. In particular:

- equal horizontal pixel steps no longer imply equal azimuth increments;
- vertical scale varies across regions and can vary between camera pairs;
- the overlap and non-overlap are governed by different projections;
- a rectangle may be filled without preserving its solid-angle support; and
- a map estimated from one scene is not a stable calibration for another scene depth.

Accordingly, the method may preserve a perceptual aspect ratio, but it does not preserve vertical field of view in the physical sense of retaining the same elevations at every azimuth.

## 6 Data Demonstration of the Local-Overlap Warp Proposal

### 6.1 Customer formulation

The customer's revised formulation is more precise than the original paper citation: it accepts that vertical field of view, object aspect ratio, rectangular output, and perfect stitching cannot all be preserved exactly. It then proposes the following priority order:

1. preserve vertical field of view;
2. preserve the aspect ratio of people and objects;
3. maintain seam alignment; and
4. push the residual geometric error primarily into the overlap regions rather than apply a cylindrical projection over the whole image.

This is a valid way to describe a non-metric display warp. It is not, however, a way to preserve calibrated 360-degree geometry. If camera interiors are protected from distortion and only the overlap bands are allowed to move, then the overlap bands must absorb nearly all of the scale, shear, and closure error. With narrow fixed-rig overlaps this makes the required local deformation very large.

## 6.2 Measured correction from local EO/IR test cases

The local test data in `C:/SVNProjects/IMU_Stabilize_v40/EO_IR_TestCases` already contain an object-based stretch analysis for paired EO and IR panoramas [13]. The EO comparison uses `Stab_false.bmp` as source and `20260730_000404_panorama_v17_f0000.bmp` as target. SIFT/object-feature affine fitting produced 3,755 inliers from 3,907 good matches with a median reprojection error of 0.621 px. The measured target/source scale was $x = 1.008511$ and $y = 1.057517$, i.e. approximately +0.85% horizontally and +5.75% vertically.

The IR comparison produced 5,123 inliers from 5,173 good matches with a median reprojection error of 0.412 px. The measured target/source scale was $x = 1.105664$ and $y = 1.177568$, i.e. approximately +10.57% horizontally and +17.76% vertically. These are not synthetic numbers; they are measured on real project images.

## 6.3 What happens if the correction is confined to overlaps

Let $S_y$ be the measured vertical correction that would otherwise be distributed across the output, let $W$ be the active panorama width, let $w_o$ be the width of one overlap band, and let $N_s$ be the number of inter-camera seams. If the camera interiors remain at vertical scale 1.0 and all correction is forced into the overlaps, the overlap fraction is

$$f = \frac{N_s w_o}{W}. \tag{7}$$

The minimum average vertical scale inside the overlap bands is then

$$\bar{s}_{\text{overlap}} = 1 + \frac{S_y - 1}{f}. \tag{8}$$

Equation (8) is optimistic because it assumes a flat scale jump inside the overlap. A visually smooth local warp that returns to scale 1.0 at both band edges requires a higher peak. For a simple raised-cosine correction profile, the approximate peak is

$$s_{\text{peak}} \approx 1 + 2\frac{S_y - 1}{f}. \tag{9}$$

**Table 4:** Overlap-only scale implied by measured project data. The EO 48.9 px row corresponds to the effective overlap in the software test demonstration; the EO 17 px and IR 29 px rows correspond to the current FPGA panorama geometries.

| Case | Measured $S_y$ | Overlap fraction | Flat overlap average | Smooth overlap peak |
|---|---|---|---|---|
| EO data, 48.9 px overlap | 1.057517 | 6.37% | 1.90× | 2.81× |
| EO data, 17 px overlap | 1.057517 | 2.21% | 3.60× | 6.20× |
| IR data, 29 px overlap | 1.177568 | 4.05% | 5.38× | 9.76× |

This calculation is the decisive data point. The customer's proposal does not remove the distortion; it spatially concentrates it. For the current EO overlap width, even the measured 5.75% vertical correction implies an average 3.60× vertical scale inside the overlap if the rest of the image is kept at 1.0×. A smooth warp profile would peak near 6.20×. The IR case is more severe. Such local gain will make lines, people, building edges, and the ground texture bend or pinch at the seam bands. It may be hidden in selected still images by choosing a favorable seam, but it is not temporally robust and it is not a calibrated ray map.

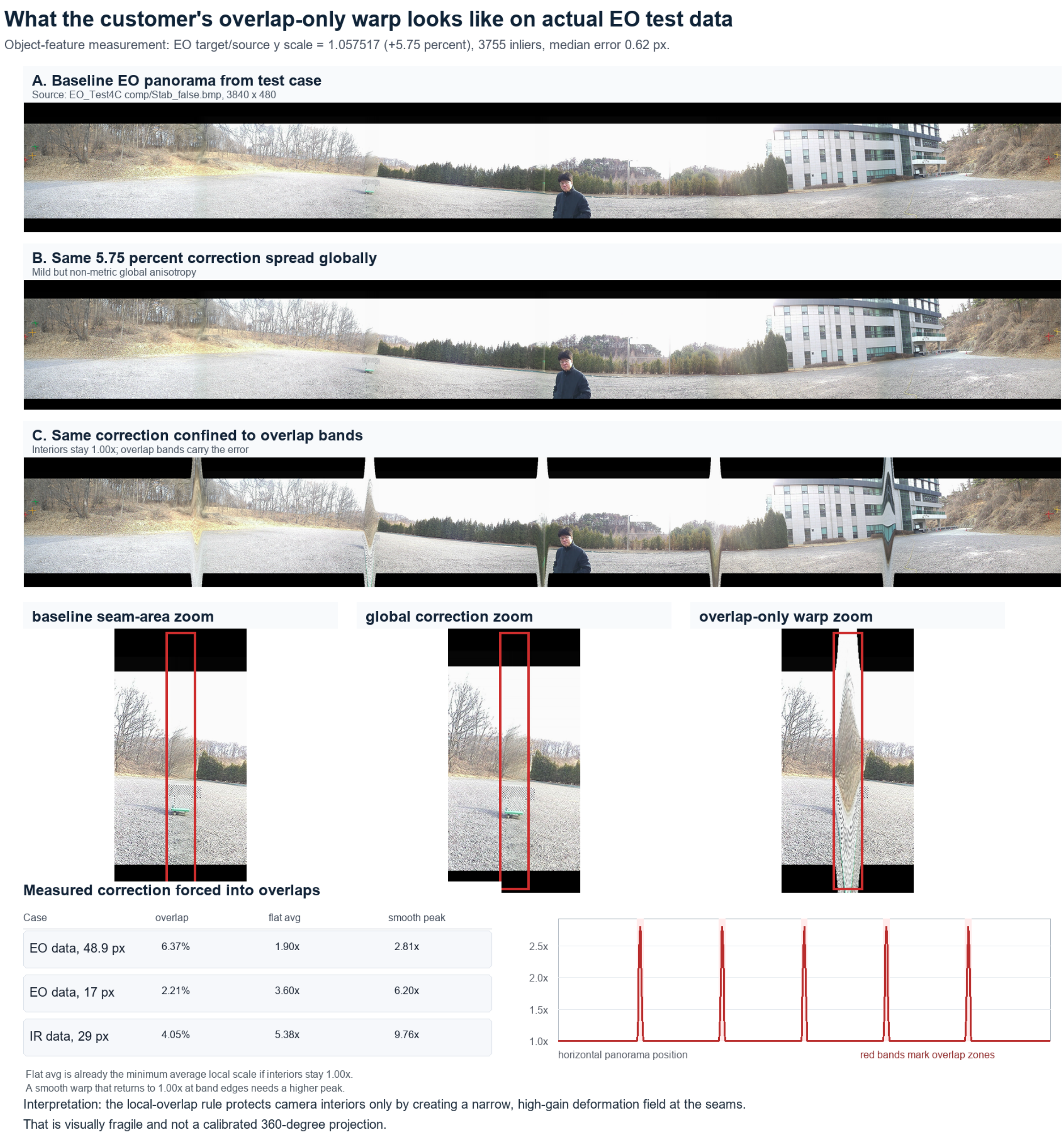


**Figure 1:** Concrete simulation of the customer-proposed overlap-only correction on actual EO test-case imagery. The same measured EO correction that is mild when spread globally becomes a high-gain deformation field when camera interiors are held fixed and the correction is confined to the overlap bands. The visual result is seam-localized stretching/pinching, not preserved calibrated panorama geometry.

## 6.4 Implication for the projection choice

The customer is correct that a non-cylindrical method can preserve more vertical source content for an operator display. The rigorous version of that idea is a calibrated multi-face or perspective-atlas representation, where each camera face keeps its own inverse ray model and the UI renders those faces on a 3-D surface. That approach does not pretend to be one rectangular cylindrical raster.

The proposed adaptive overlap warp is different. It uses a local two-dimensional deformation to make a rectangle look plausible. Because the deformation is concentrated where two cameras meet, it inherits all the problems of feature-dependent still-image stitching: depth dependence, moving-object sensitivity, seam shimmer, and loss of a stable pixel-to-ray interpretation. It can be offered as an optional view mode, but it should not replace the metric cylindrical mode used for measurement, stabilization, tracking, and any downstream analytics.

# 7 Practical Alternatives for the Existing Hardware

## 7.1 Metric flat panorama

For a single flat panorama in which every output pixel has a well-defined global ray, retain the calibrated cylindrical or equirectangular map. The output must then use one of two honest representations:

- crop to the vertical range valid at every azimuth; or
- retain the curved/scalloped validity boundary and encode unavailable regions as black or masked pixels.

This is the appropriate mode for measurement, tracking, target localization, and any downstream algorithm that depends on angular consistency.

## 7.2 Perspective atlas rendered by the UI

With the existing six cameras, the full sensor footprint can be retained as six calibrated rectilinear faces, analogous to a cubemap or a hexagonal-prism atlas. Each face preserves its source rays and may be packed into an HD transport frame. The UI then renders the faces on their corresponding 3-D surfaces.

This representation is geometrically valid as a multi-face atlas and preserves substantially more vertical sensor content. It is not one uniform cylindrical raster: horizontal angular sampling varies within each perspective face, and the UI or analytics layer must use the face-specific inverse projection. The current precomputed-map and RowRun architecture could support such a mode through different offline maps, provided row-span and memory bounds are reverified.

## 7.3 Optical overscan

If the requirement is simultaneously a full-height rectangle, one global angular projection, and no invalid top/bottom regions, additional optical coverage is required. For the EO numbers above, a vertical field near 58° is the theoretical minimum for a ±30° sector before allowing calibration tolerance, stabilization margin, lens distortion, and assembly error. A practical design would require additional margin beyond that minimum.

## 7.4 Optional non-metric operator view

A ratio-preserving, mesh-rectangled, or column-normalized warp can be offered as an optional human-view mode. Because the existing FPGA uses offline base maps and a generic fixed-point renderer, a selected non-metric warp could be precomputed and translated into the same map/RowRun representation. It should be explicitly labeled as a display projection and should not replace the calibrated mode used for measurement or machine analysis.

# 8 Assessment of the Existing G-Locked System

## 8.1 Architectural correspondence to the actual requirement

The internal C reference implementation and manuscript describe a fundamentally different control structure from the reviewed work [11, 12]:

- lens undistortion and cylindrical projection are composed offline into stable source-coordinate base maps;
- per-frame motion is supplied by the IMU rather than rediscovered from scene features;

- runtime correction is reduced to per-camera Q1.15 sine/cosine terms and a fixed-point vertical shift;
- destination samples are grouped by source-row bucket and compressed into piecewise-linear RowRuns;
- each camera renderer requires only two source lines at a time; and
- random destination writes are bounded inside an on-chip RowWindow, while DDR access remains burst-oriented.

This architecture corresponds directly to the fixed-rig, video-rate problem. The scene-independent calibration also generalizes naturally to IR, where visible-light feature networks or SIFT-based runtime matching may fail.

### 8.2 Current routed implementation evidence

The current XCKU15P routed checkpoint provides hardware evidence absent from both reviewed papers. The timing report closes the main 233.38 MHz clock with setup, hold, and pulse-width margins all positive. There are zero failing setup or hold endpoints and zero routing errors [14].

**Table 5:** Current full-design post-route snapshot. Utilization includes the complete system, including camera interfaces, DDR/MIG, HD-SDI path, control, buffers, and restored debug infrastructure; it is not a renderer-only figure.

| Resource or timing item | Used / value | Available | Utilization |
|---|---:|---:|---:|
| CLB LUTs | 48,095 | 522,720 | 9.20% |
| CLB registers | 77,034 | 1,045,440 | 7.37% |
| Block RAM tiles | 778.5 | 984 | 79.12% |
| UltraRAM | 10 | 128 | 7.81% |
| DSP48E2 | 26 | 1,968 | 1.32% |
| Worst setup slack | 0.055 ns | — | Pass |
| Worst hold slack | 0.010 ns | — | Pass |
| Worst pulse-width slack | 0.047 ns | — | Pass |
| Routable nets with errors | 0 | 137,206 routed | Pass |

The manuscript's approximately 1.93 MB value is an analytical working-set estimate for the panorama renderer's two-line caches, active RowWindow, and slice ping-pong buffers. It should not be presented as the total placed-device memory utilization. The full bitstream uses 79.12% of BRAM tiles because the complete system contains additional video, DDR, interface, buffering, and debug structures. Reporting both figures will make the hardware claim substantially more credible.

## 9 Conclusion

The reviewed literature does not justify replacing the existing calibrated cylindrical projection. Xu *et al.* propose a useful perceptual still-image warp, but their output is a piecewise, scene-dependent composition rather than a single metric projection; the total measured rate is below one image pair per second, and there is no multi-camera, video, infrared, closure, or hardware validation. Wang *et al.* solve stereoscopic disparity and naturalness using an even more elaborate iterative feature/mesh pipeline, with no cylindrical projection or runtime evidence.

The vertical-field limitation is dictated by pinhole-ray geometry. At the edge of a 60° camera sector, a 51.3°-vertical EO sensor provides approximately 45.2° of common elevation. Filling a full-height rectangle without crop or invalid pixels therefore requires either additional optical vertical coverage or a non-metric deformation. A visually pleasing rectangle is achievable; a geometrically true rectangle containing uncaptured rays is not.

For the existing hardware, the defensible engineering choices are: retain the calibrated cylindrical map for metric operation; optionally expose a masked full-footprint or multi-face perspective atlas; add optical overscan when a full-height metric rectangle is mandatory; and reserve adaptive ratio-preserving or overlap-local warps for clearly labeled operator-view modes.

**Customer-facing technical conclusion.**
The cited work performs perceptual rectangling of still-image pairs using scene-dependent two-dimensional warps. It neither preserves calibrated ray geometry nor recovers the vertical rays absent at sensor corners, and it provides no six-camera, 360-degree, video, thermal, or FPGA validation. The customer's overlap-local variant does not escape this limitation; measured EO/IR data show that it concentrates the correction into narrow seam bands, producing high local scale factors. A visually full rectangle is attainable only through non-metric rescaling; a geometrically true full-height rectangle requires additional vertical optical coverage or a masked/multi-face output representation.

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